

R Project: Real Estate Price Determinants

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1. Sworn Statement

We hereby declare, on our honour, that this report was written entirely by us, without any unauthorised external assistance. It has not been previously submitted for assessment and has never been published, in whole or in part. All passages, phrases, or ideas including tables, figures, charts, or maps that are borrowed from or refer to other sources, are fully and properly cited, without exception.

2. Introduction

We are currently living in a world where real estate prices often continue to rise in many regions, to the point where numerous households struggle to find housing that fits their financial means. This economic pressure also forces many young adults to remain living with their parents, unable to afford the cost of housing on their own.

Living in London, one of the most expensive cities in the world, where real estate prices can reach staggering levels, we have been directly confronted with this reality. This experience has led us to question the true determinants of the real estate market. In particular, we wondered why cities with similar population sizes can display significantly different housing prices. With this in mind, we decided to focus our study on the French real estate market in order to better understand regional disparities and the factors that shape these differences.

This project aims to analyze the determinants of real estate prices in France. Using data from 2022, we investigate how socio-economic factors such as median income, unemployment rate, and population density influence the price per square meter of housing. We collected data from official french government sources (Data.Gouv for real estate prices and INSEE for socio-economic indicators) to build a robust dataset comprising over 30,000 municipalities. Our approach will begin with a presentation of the selected data and variables. We will then carry out estimations with confidence intervals, followed by a goodness-of-fit test. Next, we will compare two groups using a hypothesis test, and finally conclude our analysis with a chi-square independence test.

3. Data merging

The first step of our analysis was to build our database. In fact, given that we had different data basis, we had to merge them into one. Our: INSEE provides exhaustive data for each municipality, while Data.Gouv statistics depend directly on recorded real estate transactions. As a result, some municipalities may have no available data due to the absence of property sales or purchases. This lack of representative statistics is very limited and can be considered negligible for the purposes of our study.

4. Description of data (Question 2)

```
data_final <- read_excel("Base_Finale.xlsx")

data_final$Revenu_Median <- as.numeric(data_final$Revenu_Median)
data_final$Taux_Chomage <- as.numeric(data_final$Taux_Chomage)
data_final$Prix_m2 <- as.numeric(data_final$Prix_m2)
data_final$Densite <- as.numeric(data_final$Densite)

# Sample size :
n <- nrow(data_final)
```

4.1. Definition of population and variables

The statistical units studied in this project are the **municipalities of mainland France**. The reference population consists of all municipalities present in both the DVF (sales) and INSEE (socio-economic) databases. After merging and cleaning the data, our final sample size comprises **30892** municipalities.

Our dependent variable is the **Average Real Estate Price (Prix_m2)**, expressed in Euros per square meter (€/m²). It is a quantitative continuous variable representing the average buying price of housing in a given municipality.

To explain this price, we selected four explanatory variables. The variables are summarized below:

Table 1: Summary of Variables

Variable Name	R Variable	Type	Unit	Description
Average Price	Prix_m2	Quant. Continuous	€/m ²	Target variable to explain
Median Income	Revenu_Median	Quant. Continuous	€/year	Standard of living indicator
Unemployment Rate	Taux_Chomage	Quant. Continuous	%	Economic health indicator
Density	Densite	Quant. Continuous	Inhab./km ²	Population concentration
Zone Type	Type_Zone	Categorical	N/A	Urban vs Rural classification

4.2. Description of variables

4.2.1. Average Real Estate Price

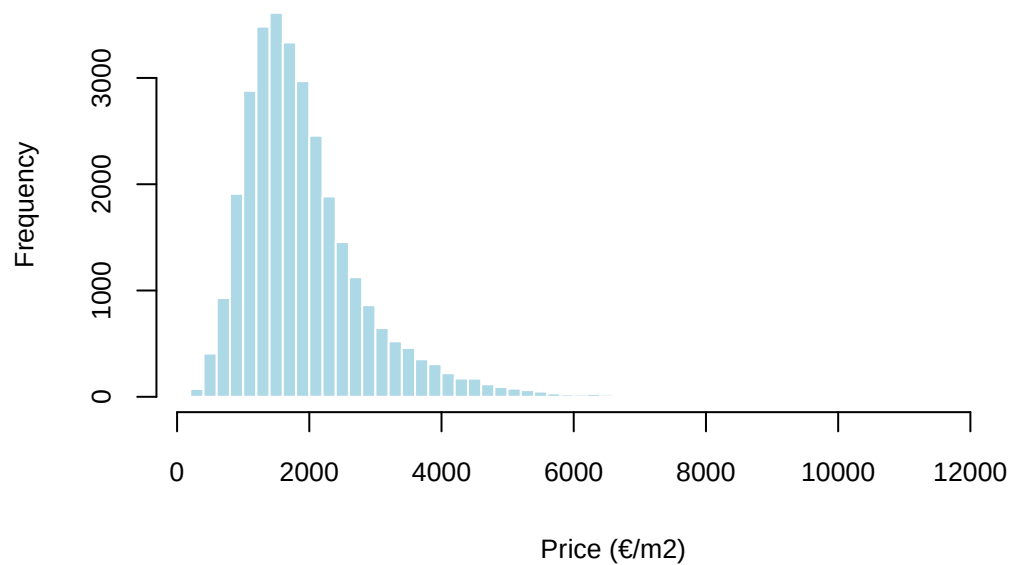
The variable Prix_m2 is quantitative continuous.

```
# Summary statistics
summary(data_final$Prix_m2)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      333   1292   1727   1935   2312   12963
```

```
# Histogram
hist(data_final$Prix_m2,
     main = "Distribution of Real Estate Prices per m²",
     xlab = "Price (€/m2)",
     col = "lightblue",
     border = "white",
     breaks = 50,
     cex.main = 0.9, # We shrank the titles and numbers
     cex.lab = 0.8,
     cex.axis = 0.8)
```

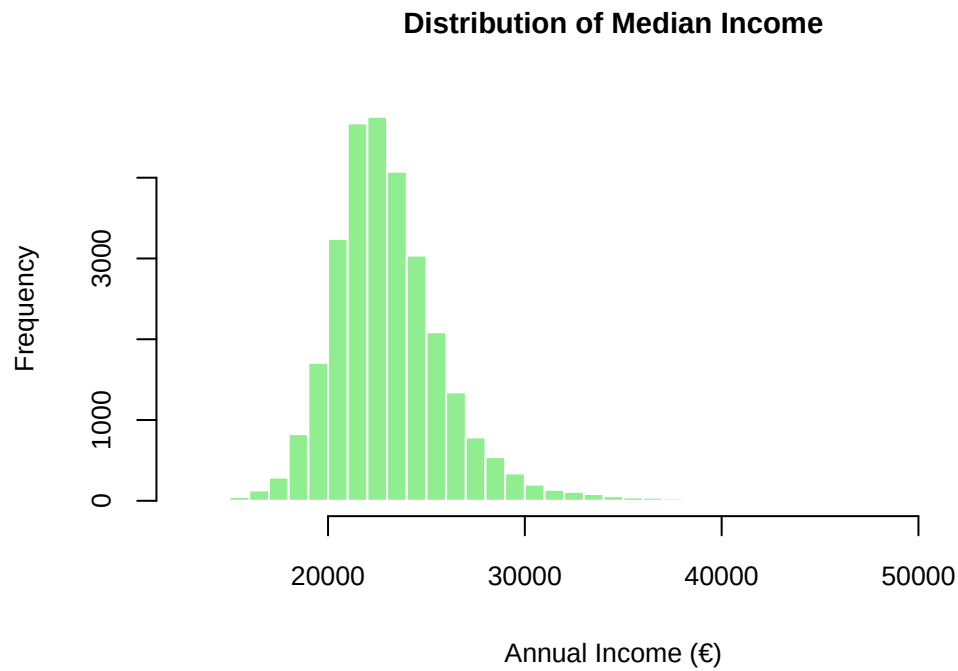
Distribution of Real Estate Prices per m²



Comment: The distribution is right-skewed. The majority of municipalities have prices between 1,000 and 3,000 €/m². High values correspond to major urban centers and exclusive coastal areas.

4.2.2. Median Income (Revenu_Median)

```
hist(data_final$Revenu_Median,  
      main = "Distribution of Median Income",  
      xlab = "Annual Income (€)",  
      col = "lightgreen",  
      border = "white",  
      breaks = 40,  
      cex.main = 0.9,  
      cex.lab = 0.8,  
      cex.axis = 0.8)
```

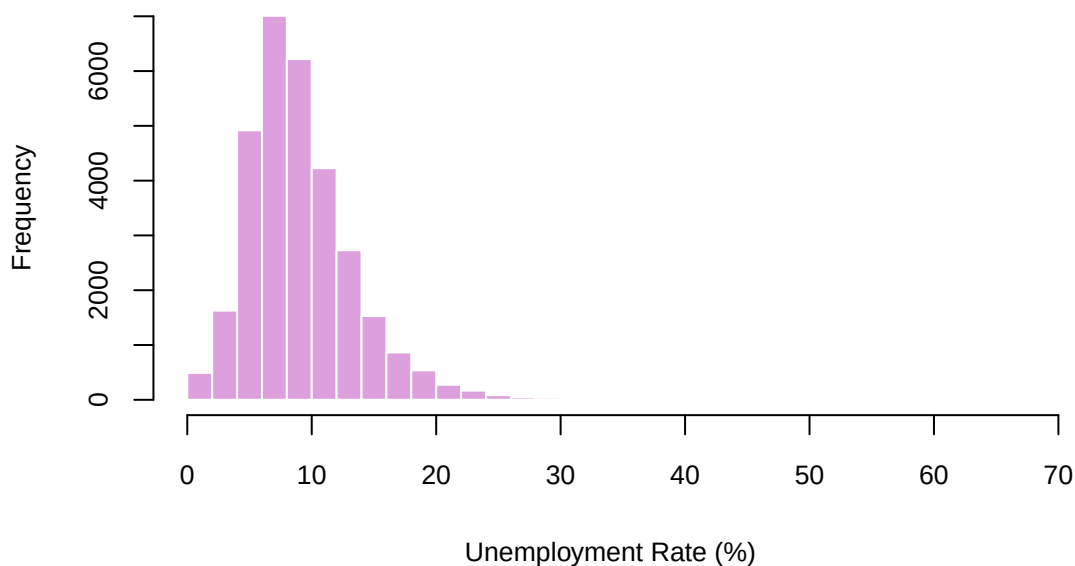


Comment: The income distribution is bell-shaped and centered around 22,000 €, suggesting a normal distribution.

4.2.3. Unemployment Rate (Taux_Chomage)

```
hist(data_final$Taux_Chomage,  
      main = "Distribution of Unemployment Rate",  
      xlab = "Unemployment Rate (%)",  
      col = "plum",  
      border = "white",  
      breaks = 30,  
      cex.main = 0.9,  
      cex.lab = 0.8,  
      cex.axis = 0.8)
```

Distribution of Unemployment Rate



Comment: Most municipalities have an unemployment rate between 5% and 10%, with a few areas suffering from significantly higher rates.

4.2.4. Zone Type (Type_Zone)

This categorical variable classifies municipalities as “Urban” or “Rural”.

```
# Frequency table
tab_zone <- table(data_final$Type_Zone)
prop_zone <- prop.table(tab_zone)

# Formatting the table
resume_zone <- rbind(Count = tab_zone, Frequency = round(prop_zone, 2))

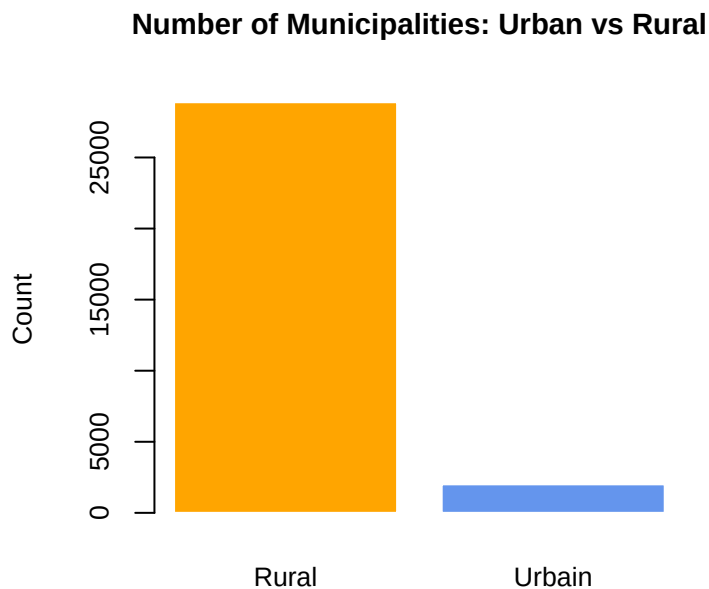
kable(resume_zone, caption = "Distribution of Municipalities by Zone Type") %>%
  kable_styling(latex_options = "hold_position")
```

Table 2: Distribution of Municipalities by Zone Type

	Rural	Urban
Count	28887.00	2005.00
Frequency	0.94	0.06

```
barplot(tab_zone,
  main = "Number of Municipalities: Urban vs Rural",
  ylab = "Count",
```

```
col = c("orange", "cornflowerblue"),
border = "white",
cex.main = 0.9,
cex.names = 0.8,
cex.lab = 0.8,
cex.axis = 0.8)
```



Comment: The dataset contains a larger number of rural municipalities, reflecting the administrative fragmentation of the French territory.

5. Point and Interval Estimation (Question 3)

We aim to estimate the unknown population parameter μ , which corresponds to the **true mean price per square meter** of all French municipalities.

5.1. Point Estimation

We use the sample mean $\bar{x}$ as an unbiased point estimator for μ .

```
# Calculation of the sample mean (Point Estimate)
mean_est <- mean(data_final$Prix_m2)

# Calculation of standard deviation (s) and sample size (n)
sd_est <- sd(data_final$Prix_m2)
```

```
# Checking
n <- nrow(data_final)

# Display the mean
mean_est
```

```
## [1] 1935.296
```

Result: The point estimate for the average real estate price in France is **1935.3 €/m²**.

5.2. Confidence Interval Estimation

Since our sample size is extremely large ($n = 30892 > 30$), we apply the **Central Limit Theorem**. We can approximate the distribution of the mean as Normal, even if the underlying data is not perfectly normal.

The formula for the confidence interval is:

$$IC_{1-\alpha} = \left[\bar{x} - z_{1-\alpha/2} \frac{s}{\sqrt{n}} ; \bar{x} + z_{1-\alpha/2} \frac{s}{\sqrt{n}} \right]$$

```
# Calculation of the standard error
standard_error <- sd_est / sqrt(n)

# 90% Confidence Interval (alpha = 10%)
alpha_90 <- 0.10
z_90 <- qnorm(1 - alpha_90 / 2) # Critical value (approx 1.645)

lower_90 <- mean_est - z_90 * standard_error
upper_90 <- mean_est + z_90 * standard_error

# 95% Confidence Interval (alpha = 5%)
alpha_95 <- 0.05
z_95 <- qnorm(1 - alpha_95 / 2) # Critical value (approx 1.96)

lower_95 <- mean_est - z_95 * standard_error
upper_95 <- mean_est + z_95 * standard_error

# We need to format results for display
res_CI <- data.frame(
  Level = c("90%", "95%"),
  Lower_Bound = round(c(lower_90, lower_95), 2),
  Upper_Bound = round(c(upper_90, upper_95), 2),
  Margin_Error = round(c(z_90 * standard_error, z_95 * standard_error), 2)
)

# Display with Kable
kable(res_CI, caption = "Confidence Intervals for Mean Price") %>%
  kable_styling(latex_options = "hold_position")
```

Table 3: Confidence Intervals for Mean Price

Level	Lower_Bound	Upper_Bound	Margin_Error
90%	1926.04	1944.55	9.26
95%	1924.26	1946.33	11.03

Interpretation:

- **At the 90% confidence level**, we estimate that the true average price in France lies between **[1926.04 ; 1944.55] €/m²**.
- **At the 95% confidence level**, the interval widens slightly to **[1924.26 ; 1946.33] €/m²**.

Comment on Precision: The intervals are extremely narrow (the margin of error is very small, less than 20€). This high precision is a direct result of our massive sample size ($n \approx 30,000$). Mathematically, as n increases, the standard error $\frac{s}{\sqrt{n}}$ tends toward 0, shrinking the confidence interval.

6. Conformity Test (Question 4)

We conduct a conformity test (one-sample t-test) to check if our sample mean differs significantly from a theoretical value.

Scenario: Let us assume that a national real estate agency claims that the average price per m^2 in France is **2,500 €**. We want to test this claim against our data.

We define our hypotheses as follows:

$$\begin{cases} H_0 : \mu = 2500 & \text{(The true mean is equal to 2,500 €)} \\ H_1 : \mu \neq 2500 & \text{(The true mean is different from 2,500 €)} \end{cases}$$

6.1. Test at 5% Significance Level ($\alpha = 0.05$)

```
# Theoretical mean we want to test
mu_0 <- 2500

# T-test (two-sided) with 95% confidence level
test_5 <- t.test(data_final$Prix_m2, mu = mu_0, alternative = "two.sided", conf.level = 0.95)

# We build a table in order to present the results better
res_test_5 <- data.frame(
  "Statistic_t" = round(test_5$statistic, 2),
  "P_Value" = format.pval(test_5$p.value, digits = 3, eps = 0.001),
  "Lower_Bound" = round(test_5$conf.int[1], 2),
  "Upper_Bound" = round(test_5$conf.int[2], 2),
  "Mean_Estimate" = round(test_5$estimate, 2)
)

# Display result (we use Kable)
kable(res_test_5, caption = "T-Test Results (Alpha = 5\\%)") %>%
  kable_styling(latex_options = "hold_position")
```

Table 4: T-Test Results (Alpha = 5%)

	Statistic_t	P_Value	Lower_Bound	Upper_Bound	Mean_Estimate
t	-100.32	<0.001	1924.26	1946.33	1935.3

Decision Rule: Since the p-value (<0.001) is strictly lower than $\alpha = 0.05$, we **reject the Null Hypothesis** H_0 . We have sufficient evidence to state that the average price is statistically different from 2,500 €. The 95% confidence interval is [1924.26; 1946.33], which does not contain 2,500.

6.2. Test at 10% Significance Level ($\alpha = 0.10$)

```
# T-test with 90% confidence level
test_10 <- t.test(data_final$Prix_m2, mu = mu_0, alternative = "two.sided", conf.level = 0.90)

# Again we build a table in order to present the results better
res_test_10 <- data.frame(
  "Statistic_t" = round(test_10$statistic, 2),
  "P_Value" = format.pval(test_10$p.value, digits = 3, eps = 0.001),
  "Lower_Bound" = round(test_10$conf.int[1], 2),
  "Upper_Bound" = round(test_10$conf.int[2], 2),
  "Mean_Estimate" = round(test_10$estimate, 2)
)

# Display result
kable(res_test_10, caption = "T-Test Results (Alpha = 10\\%)") %>%
  kable_styling(latex_options = "hold_position")
```

Table 5: T-Test Results (Alpha = 10%)

	Statistic_t	P_Value	Lower_Bound	Upper_Bound	Mean_Estimate
t	-100.32	<0.001	1926.04	1944.56	1935.3

Decision Rule: Since the p-value (<0.001) is also lower than $\alpha = 0.10$, we **reject** H_0 at the 10% significance level as well. The 90% confidence interval is [1926.04; 1944.56], which is narrower but still does not include the value 2,500.

Global Conclusion on Conformity: Regardless of the risk level (5% or 10%), our data clearly contradicts the hypothetical claim of 2,500 €/m². The actual average price (approx. 1,935 €) is significantly lower than this reference value.

7. Comparison of Two Subgroups (Question 5)

We wish to compare the average real estate prices between two specific subgroups of our population: **Urban** municipalities and **Rural** municipalities.

We define two independent samples: * Sample 1: Prices in **Urban** areas (n_1, μ_1, σ_1^2) * Sample 2: Prices in **Rural** areas (n_2, μ_2, σ_2^2)

Step 1: Test for Equality of Variances Before comparing the means, we must check if the variances of the two groups are equal.

$$\begin{cases} H_0 : \sigma_1^2 = \sigma_2^2 & (\text{Variances are equal}) \\ H_1 : \sigma_1^2 \neq \sigma_2^2 & (\text{Variances are different}) \end{cases}$$

```
# Extracting data
prices_urban <- data_final$Prix_m2[data_final$Type_Zone == "Urbain"]
prices_rural <- data_final$Prix_m2[data_final$Type_Zone == "Rural"]

# Test for variances
var_test <- var.test(prices_urban, prices_rural, alternative = "two.sided")

res_var <- data.frame(
  "Ratio_of_variances" = round(var_test$estimate, 2),
  "P_Value" = format.pval(var_test$p.value, digits = 3, eps = 0.001),
  "Conf_Int_Low" = round(var_test$conf.int[1], 2),
  "Conf_Int_High" = round(var_test$conf.int[2], 2)
)

kable(res_var, caption = "Equality of Variances Test Results") %>%
  kable_styling(latex_options = "hold_position")
```

Table 6: Equality of Variances Test Results

	Ratio_of_variances	P_Value	Conf_Int_Low	Conf_Int_High
ratio of variances	2.38	<0.001	2.23	2.54

Conclusion on Variances: Since the p-value is extremely low (< 0.05), we reject the null hypothesis. The variances are **not equal**. Consequently, we will perform a **Two-sample t-test** (comparison of means) which does not assume equal variances in order to compare the means.

Step 2: Comparison of Means (Two-sample t-test)

We test if there is a significant difference between the average price in Urban areas and in Rural areas.

$$\begin{cases} H_0 : \mu_1 = \mu_2 & (\text{Average prices are identical}) \\ H_1 : \mu_1 \neq \mu_2 & (\text{Average prices are different}) \end{cases}$$

7.1. Test at 5% Significance Level ($\alpha = 0.05$)

```
# We use 'two.sided' to see if there is a difference, and get the CI of that difference
ttest_5 <- t.test(prices_urban, prices_rural, alternative = "two.sided", var.equal = FALSE, conf.level = 0.95)

# here we format the results
res_ttest_5 <- data.frame(
  "Difference_Mean" = round(ttest_5$estimate[1] - ttest_5$estimate[2], 2),
  "Statistic_t" = round(ttest_5$statistic, 2),
  "P_Value" = format.pval(ttest_5$p.value, digits = 3, eps = 0.001),
```

```

"Lower_Bound_Diff" = round(ttest_5$conf.int[1], 2),
"Upper_Bound_Diff" = round(ttest_5$conf.int[2], 2)
)

kable(res_ttest_5, caption = "Comparison of Means Results (Alpha = 5\\%)" %>%
  kable_styling(latex_options = "hold_position")

```

Table 7: Comparison of Means Results (Alpha = 5%)

	Difference_Mean	Statistic_t	P_Value	Lower_Bound_Diff	Upper_Bound_Diff
mean of x	1082.95	33.97	<0.001	1020.42	1145.47

Decision: The p-value is practically zero. We **reject** H_0 . There is a significant difference between urban and rural prices. At 95% confidence, Urban areas are on average between **1020.42 €/m²** and **1145.47 €/m²** more expensive than Rural areas.

7.2. Test at 10% Significance Level ($\alpha = 0.10$)

```

# Same test with 90% confidence level
ttest_10 <- t.test(prices_urban, prices_rural, alternative = "two.sided", var.equal = FALSE, conf.level

# Formatting results
res_ttest_10 <- data.frame(
  "Difference_Mean" = round(ttest_10$estimate[1] - ttest_10$estimate[2], 2),
  "Statistic_t" = round(ttest_10$statistic, 2),
  "P_Value" = format.pval(ttest_10$p.value, digits = 3, eps = 0.001),
  "Lower_Bound_Diff" = round(ttest_10$conf.int[1], 2),
  "Upper_Bound_Diff" = round(ttest_10$conf.int[2], 2)
)

kable(res_ttest_10, caption = "Comparison of Means Results (Alpha = 10\\%)" %>%
  kable_styling(latex_options = "hold_position")

```

Table 8: Comparison of Means Results (Alpha = 10%)

	Difference_Mean	Statistic_t	P_Value	Lower_Bound_Diff	Upper_Bound_Diff
mean of x	1082.95	33.97	<0.001	1030.48	1135.41

Global Comment: The results are consistent for both significance levels. The p-value remains very small, confirming that **Urban municipalities are significantly more expensive than Rural ones**. The positive difference observed in the confidence intervals confirms the “Urban Premium” in the real estate market.

8. Chi-square Test of Independence (Question 6)

We wish to check if there is a dependency link between **Real Estate Prices** and the **Unemployment Rate**. Since both variables are quantitative, we first need to separate them into categorical classes (low, medium, high) to perform a Chi-square (χ^2) test of independence.

We define our hypotheses:

$$\begin{cases} H_0 : \text{The variables are independent} \\ H_1 : \text{The variables are dependent (associated)} \end{cases}$$

8.1. Data Preparation and Contingency Table

We divide both variables into 3 equal classes based on their quantiles (tertiles).

```
# Dividing prices into 3 classes (low, medium, high):
breaks_price <- quantile(data_final$Prix_m2, probs = c(0, 1/3, 2/3, 1), na.rm = TRUE)
data_final$Price_Class <- cut(data_final$Prix_m2, breaks = breaks_price,
                             include.lowest = TRUE, labels = c("Low", "Medium", "High"))

# Dividing unemployment into 3 classes (low, medium, high):
breaks_unemp <- quantile(data_final$Taux_Chomage, probs = c(0, 1/3, 2/3, 1), na.rm = TRUE)
data_final$Unemp_Class <- cut(data_final$Taux_Chomage, breaks = breaks_unemp,
                             include.lowest = TRUE, labels = c("Low", "Medium", "High"))

# We create a contingency table (observed frequencies)
table_obs <- table(data_final$Unemp_Class, data_final$Price_Class)

# Displaying the table
kable(addmargins(table_obs), caption = "Contingency Table: Unemployment vs Price") %>%
  kable_styling(latex_options = "hold_position")
```

Table 9: Contingency Table: Unemployment vs Price

	Low	Medium	High	Sum
Low	2862	3472	3943	10277
Medium	3079	3472	3726	10277
High	4326	3338	2611	10275
Sum	10267	10282	10280	30829

8.2. Conducting the Test

We perform the independance test at 5% and 10% significance levels.

```
# the test :
chisq_res <- chisq.test(table_obs)

# summary for the result :
res_chisq <- data.frame(
  "Chi_Square_Stat" = round(chisq_res$statistic, 2),
  "Degrees_of_Freedom" = chisq_res$parameter,
  "P_Value" = format.pval(chisq_res$p.value, digits = 3, eps = 0.001)
)

# Displaying results
kable(res_chisq, caption = "Chi-Square Test Results") %>%
  kable_styling(latex_options = "hold_position")
```

Table 10: Chi-Square Test Results

	Chi_Square_Stat	Degrees_of_Freedom	P_Value
X-squared	666.44	4	<0.001

Interpretation and Conclusion:

- **P-value Analysis:** The p-value is extremely low (<0.001), well below both thresholds ($\alpha = 0.05$ and $\alpha = 0.10$).
- **Decision:** We **reject the Null Hypothesis** (H_0).
- **Comment:** There is a significant dependence between the Unemployment Rate and Real Estate Prices. These two economic variables are not independent; areas with high unemployment tends to have different pricing structures compared to areas with low unemployment.

8.3. Visualization of the Dependence

To visualize this link, we use a plot. The width of the columns represents the proportion of unemployment classes, and the colors represent the distribution of prices within those classes.

```
plot(table_obs,
     main = "Plot: Unemployment vs Price",
     xlab = "Unemployment Level",
     ylab = "Price Level",
     col = c("lightblue", "cornflowerblue", "darkblue"),
     border = "white")
```

Plot: Unemployment vs Price

